GENESIS Master Developer Bible

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1 Purpose and authority

GENESIS is an original seven-era evolution sandbox inspired by the creativity, personality, and colorful presentation of Spore (2008). The player shapes a lineage from small aquatic life into a civilization capable of exploring space. Earlier decisions remain relevant as the scale expands. This manual defines the intended experience, system boundaries, production workflows, and evidence agents must deliver.

Status: This is a new design baseline for owner review, not a recovered copy of the earlier chat Bible. The confirmed requirements below are binding. Era names, detailed mechanics, numerical values, technical structures, and milestone scopes elsewhere in this document are proposals until approved. Agents may investigate them and build isolated, reversible prototypes; they may not treat proposals as permission to expand production scope. No screenshots, concept art, or earlier visual boards were available when this version was written. The visual standards here are written specifications, not approved visual references.

Confirmed requirements

- Use Godot 4, Blender, and supervised AI coding agents.
- Maintain seven evolutionary eras that build on prior gameplay rather than discard it.
- Creature anatomy must meaningfully influence gameplay.
- Author and finalize every production 3D graphical asset in Blender.
- Target cohesive, colorful, stylized visual quality comparable to Spore (2008), using original designs.
- Agents must not independently change the vision, introduce incompatible systems, or expand scope without approval.
- Prioritize working gameplay, meaningful automated tests, playable Windows builds, and incremental development.
- Keep Genesis entirely separate from Orbit, including repositories, assets, saves, branding, and project configuration.

Decision hierarchy

Owner decisions override this Bible. Approved Bible revisions override implementation notes. Approved feature specifications refine the Bible without contradicting it. Code is evidence of current behavior, not authority to redefine the design. When two sources conflict, record the conflict and continue independent work that does not depend on resolving it.

Use three labels in all specifications: **CONFIRMED** for owner requirements, **PROPOSED** for unapproved design, and **IMPLEMENTED** only when a specific build and verification evidence exist. Approval does not imply implementation; implementation does not imply approval.

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2 Experience pillars

A creature the player recognizes as their own. Shape, movement, behavior, and history distinguish a lineage. Give organisms expressive eyes, readable poses, and visible reactions without copying existing characters.

Evolution through understandable tradeoffs. Every meaningful body change creates capabilities and costs. Speed, protection, feeding reach, senses, stamina, and social expression interact. A single expensive part must not solve every ecological problem.

Continuity across scales. Expanding from organism to population and civilization adds management options while retaining habitats, species relationships, ancestry, and embodied play. Simulation detail can change with distance; historical identity cannot silently reset.

Stories from interacting systems. A drought changes migration, a migration changes neighbors, and a former rival may become a trading partner. Prefer consequences players can observe over hidden random punishment.

Creation integrated with play. Editors affect functioning organisms, settlements, vehicles, and habitats. Players can inspect why a design succeeds or fails before committing.

Explicit exclusions from the initial production scope

No multiplayer, live service, blockchain, user-generated scripting, paid asset marketplace, unrestricted universe simulation, or runtime language-model dependency is included. These require separate owner approval. Do not copy Spore assets, names, UI layouts, audio, or proprietary systems. Its aesthetic is a comparison target, not a source asset library.

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3 Core loop and player agency

The recurring loop is **observe → adapt → act → experience consequences → preserve discoveries → expand**. The player observes environmental pressure, chooses a body or behavioral response, tests it through play, and carries the result into their lineage history.

Separate three progression resources. **Energy** sustains the current organism. **Adaptation potential** budgets inherited changes. **Knowledge** records discoveries and unlocks cultural or technological possibilities. Resource names and tuning are proposed. Keep sources and sinks distinct so eating does not automatically buy every civilization technology.

Direct control remains available where technically feasible after population-level systems unlock. Strategic views issue intentions; local play reveals consequences. Later-era embodied play must have useful activities rather than existing only as a camera mode.

A new run selects a seed and accessible difficulty preset, introduces movement and feeding through safe play, and creates a persistent lineage ID. Present a clear next opportunity without requiring a lengthy opening cinematic. The first playable milestone should prove movement, feeding, threat response, adaptation, and save continuity before adding expansive world content.

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4 Seven era progression proposal

These era boundaries and names require owner approval. Their purpose is to prevent disconnected minigames and define exactly what persists.

Era	Primary play	New layer	Retained systems
1 Microscopic	Swim, feed, evade, reproduce	Basic body editor and ecological niches	Founding genome, seed, ancestry
2 Aquatic	Explore depth zones and hunt or forage	Multicellular locomotion and richer senses	Diet, body tradeoffs, species relations
3 Terrestrial	Survive shore and land habitats	Limbs, stamina, nests, migration	Aquatic habitats and swimming adaptations
4 Social	Coordinate groups and build relationships	Group roles, communication, learned behavior	Individual anatomy, ecology, ancestry
5 Tribal	Manage a settlement and surrounding territory	Tools, construction, shared resources	Groups, local creatures, habitats, rivals
6 Civilization	Coordinate settlements and infrastructure	Research, diplomacy, transport, regional policy	Tribal histories, biological needs, ecology
7 Interstellar	Explore and establish distant habitats	Spacecraft, colonies, cross-world exchange	Homeworld, lineage, culture, relationships

Transition contract

A transition requires demonstrated capability, a survivable destination, and a player choice. Show what unlocks, which responsibilities change, and what remains available. Save before transition. Retain the prior save as a recoverable checkpoint. The same lineage and world records continue after transition.

Never force a land-adapted build onto a fully aquatic lineage merely because a timer expired. Offer an accessible evolutionary route, allow continued aquatic play within supported scope, and explain any era gate. Alternative paths require bounded content plans; seven eras do not authorize seven full campaigns per body plan.

Every era specification must identify a local activity from the earlier era that remains playable, one earlier adaptation that still matters, and one historical relationship that persists. A transition fails acceptance if any of these is replaced by a cosmetic badge alone.

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5 Microscopic era

Loop: search for food, manage exposure, avoid predators, earn adaptation potential, reproduce, and edit the next generation. Start in a readable shallow environment with food and threats that communicate their roles through silhouette and motion.

Use a three-dimensional presentation with a constrained movement plane for the first prototype. This reduces control complexity without making production 3D graphics exempt from the Blender rule. The exact camera and movement constraint are proposed and should be tested early.

Feeding uses mouth capability, target compatibility, distance, and cooldown. Herbivorous, predatory, and mixed diets are supported design directions. Different food distributions encourage movement. A defensive part discourages particular attacks; it does not create universal invulnerability.

Reproduction opens a bounded editor at a safe opportunity. Validate the edited organism before spending adaptation potential. Preview changes in speed, turning, feeding, defense, and energy demand. Death returns to a fair lineage checkpoint with clear feedback; do not permanently erase a run by default.

Acceptance: a player can complete two generations, distinguish food from danger, make a change with an observable benefit and cost, save and reload, and recognize their lineage. Test starvation recovery, invalid edits, repeat death, and food exhaustion. No next-era work is required to demonstrate this loop.

6 Aquatic era

Loop: navigate habitat zones, find suitable food, compete or cooperate with species, manage energy, and evolve locomotion or sensing. Introduce depth gradually so navigation and threat readability remain manageable.

Fins, tails, buoyant structures, sensory organs, and body proportions influence movement through shared anatomy rules. Depth hazards must communicate danger before causing irreversible loss. Add oxygen or pressure only if they create understandable choices and fit the approved scope.

Species occupy niches rather than uniformly spawning around the player. Shelters and feeding grounds support repeated encounters. Nests or spawning sites create recognizable locations. Larger organisms can exist as environmental threats before becoming fully simulated creatures.

Continuity: retain genome identity, dietary capabilities, earlier species records, and ancestry. Scale the depiction of microscopic life into the aquatic ecosystem rather than deleting it from history. **Acceptance:** compare two body variants in the same route, verify different movement outcomes, and demonstrate that discovery and relationships survive a reload.

7 Terrestrial era

Loop: explore shorelines, locate shelter and food, navigate terrain, manage stamina, establish a nest, and adapt to environmental pressure. Shore zones connect swimming and walking so the transition has visible meaning.

Support a small approved set of body plans first. Limb placement influences support, reach, gait, and turning, while the game constrains impossible structures with clear editor feedback. Avoid promising physically accurate locomotion for arbitrary anatomy.

Water remains useful and accessible. Swimming organs may offer river access while competing with land efficiency. Terrain creates choices through slope, vegetation, cover, and exposure. Use visible habitat conditions before adding complex climate simulation.

Acceptance: an aquatic adaptation remains useful, an edited land body crosses terrain differently, a nest persists, and a predator can investigate, pursue, disengage, and recover without getting stuck. Demonstrate a safe recovery route if the player enters unsuitable terrain.

8 Social era

Loop: meet individuals, communicate intentions, recruit companions, coordinate food or defense, and pass learned behavior between generations. Social success is an alternative supported strategy, not a universal replacement for survival.

Individuals have identity, needs, memory of significant encounters, and bounded roles such as scout, defender, or forager. Communication uses readable signals and responses. Anatomy can influence range or modality, but sensory differences must have accessible substitutes so a body choice does not make controls unusable.

Relationships combine recent encounters with persistent reputation. A single accidental collision should not cause permanent war. Group orders include follow, wait, investigate, retreat, and assist. Companions should clearly acknowledge orders and report inability to comply.

Acceptance: recruit and dismiss a companion, complete a group task, demonstrate refusal or retreat, and reload with identities and significant relationships intact. Test loss of the leader, unreachable orders, and separation across streaming boundaries.

9 Tribal era

Loop: gather resources, assign group work, construct useful structures, negotiate with neighbors, and sustain a settlement. Embodied play retains exploration, feeding, observation, and meaningful local interactions.

Introduce tools as extensions of anatomy. Manipulation ability influences tool interaction; alternate approved body plans need equivalent viable paths. Do not simply prohibit advancement for creatures without human hands.

Construction uses functional modules, placement previews, collision checks, access validation, and explicit resource costs. Settlement tasks run through a shared reservation system to prevent several workers collecting the same item. Pause and inspect queues. Construction cancellation refunds according to an explicit policy.

Diplomacy supports cooperation, exchange, territorial disagreement, and conflict. Cultural choices emerge from repeated actions; do not reduce every society to one irreversible morality score. **Acceptance:** establish a small functioning settlement, resolve a resource shortage, interact with a neighboring group, and verify individual lineage and body traits remain relevant.

10 Civilization era

Loop: connect settlements, manage production and transport, research capabilities, negotiate regional relationships, and respond to ecological consequences. Begin with a bounded region rather than a fully simulated planet.

Research requires prerequisites and practical inputs. Transport depends on routes and actual delivery state. Prevent silent teleportation of scarce goods unless an abstraction is explicitly approved and disclosed in the UI. Settlements inherit histories and biological needs from their founders.

Economic, diplomatic, and military options should each solve some problems with associated costs. Conflict targets and consequences must be readable. Ecosystem pressure may affect food, migration, or land use through coarse regional models; do not run every organism at full fidelity offscreen.

Acceptance: two settlements exchange resources, a disrupted route has a visible consequence, one diplomatic agreement changes behavior, and a local creature interaction remains playable. Show that earlier relationships and environmental records survive regional simulation.

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11 Interstellar era

Loop: prepare expeditions, travel, survey habitats, establish viable outposts, exchange resources and knowledge, and revisit home. Start with a small authored star region and seeded destinations. Infinite content is not a milestone requirement.

Ships have functional constraints such as range, capacity, and habitat support. A ship editor should use approved modules before supporting arbitrary forms. Travel presentation, seamless transitions, and planetary scale need separate technical decisions; no Orbit-specific travel requirement is inherited.

Colonization depends on species needs and local conditions. Preserve homeworld relevance through culture, resources, relationships, and return activities. New worlds contain discoverable ecosystems with bounded simulation. First contact relies on readable goals and memory of encounters.

Acceptance: visit a second world, establish an outpost, return home, and verify the same lineage, culture, discoveries, and relationships. Test interrupted travel and unavailable destinations without corrupting saves.

12 Anatomy and evolution model

Represent the genome as a versioned graph of body nodes, attachments, proportions, appearance parameters, and inherited traits. Stable IDs identify parts independently of scene paths. Separate the genome from the derived phenotype and the current actor state.

Genome stores inheritable design. **Phenotype** derives capabilities from anatomy. **Actor state** stores current health, energy, stamina, injuries, and temporary effects. **Lineage history** records parent generation, major edits, discoveries, and era transitions.

Anatomy feature	Gameplay influence	Required balancing cost
Mouth and feeding apparatus	Food compatibility and interaction reach	Energy demand or handling time
Fins and tails	Swimming propulsion and steering	Drag, mass, or land inefficiency
Legs and feet	Land support, stride, terrain access	Energy use and structural budget
Armor and spines	Damage response and deterrence	Mass, maneuverability, maintenance
Eyes and sensory organs	Detection coverage and information	Budget, exposure, or energy
Grasping structures	Carrying, tools, construction tasks	Mass and specialization
Communication organs	Signal modality and range	Budget and environmental limitations

Calculate derived stats through a single authoritative service with documented units and caps. Avoid granting full bonuses independently per duplicate part. Use diminishing returns and valid attachment rules. Numerical curves are tuning data, not scattered constants.

The editor must support undo, redo, mirroring, removal, rotation, scale limits, cost preview, and a test space. Committing is transactional: validate, derive phenotype, confirm affordability, write genome, then instantiate. Failed validation preserves the previous valid creature and resource balance.

For modular animation, map supported body plans to documented rig families and gait templates. Unusual designs receive clear constraints or approved fallback motion. Runtime deformation may operate on Blender-authored meshes; it cannot become an alternate pipeline for bypassing Blender source requirements. Validate collision proxies after deformation.

13 Ecology and simulation

Use environmental regions with carrying capacity, food regeneration, habitat suitability, and species presence. Local organisms use detailed AI. Distant populations use aggregate updates. Transfer between representations conserves identities where required and prevents duplication of resources or creatures.

Seed world generation, encounter selection, and test scenarios separately. Record relevant seeds in bug reports. Deterministic subsystem tests are a target; bit-for-bit physics determinism across machines is not assumed.

AI uses explicit states such as idle, forage, investigate, approach, feed, flee, pursue, socialize, rest, and recover. Perception determines available information; decisions use that information rather than omniscient player coordinates. Use line of sight, range, obstruction, and bounded memory as appropriate.

Every pursuit has disengagement rules. Every navigation task has timeout and recovery behavior. Schedule perception and decisions at bounded rates, with close actors receiving greater fidelity. Pool effects only when profiling justifies it. Simulate depletion and regeneration to prevent both endless free resources and unrecoverable empty worlds.

14 Interaction and conflict

One interaction resolver governs eligible targets, distance, visibility, required anatomy, cost, cooldown, and effect. UI previews and actual actions query the same rules. Failed actions explain the missing requirement without consuming resources.

Combat emphasizes readable intent, windup, contact, reaction, and recovery. Use locomotion and positioning alongside anatomy. Attack visuals and damage windows must agree. Friendly fire, death penalties, injuries, and encounter difficulty remain explicit tuning decisions.

Social interaction uses recognition, signal, response, and consequence. Avoid a universal rhythm minigame unless separately approved. Trade and diplomacy show exact offers and commitments before execution. Actions with lasting consequences need confirmation or a meaningful undo policy.

15 Art direction and visual acceptance

Aim for rounded organic forms, expressive creatures, bold silhouettes, clear color grouping, and inviting alien landscapes. Spore (2008) is the fidelity benchmark. Originality applies to anatomy, architecture, plants, vehicles, symbols, UI, and sound.

Use broad readable materials with controlled surface detail. Avoid photoreal texture noise, arbitrary kitbash mixtures, and inconsistent polygon language. A creature should remain identifiable at gameplay camera distance. Eyes, face orientation, and body pose should communicate attention and emotion.

Palette proposal: rich greens and turquoise habitats, warm mineral accents, soft cream highlights, and cool blue shadows. This is not an approved color board and does not inherit Orbit's palette. Establish biome-specific palettes with shared saturation and lighting discipline.

Lighting should reveal forms and interactive targets. Use soft atmospheric depth, controlled bloom, readable shadows, and restrained fog. Color grading cannot hide poor material consistency. Renderer choice and effect availability must be verified in the pinned Godot version; do not assume every effect works in every rendering backend.

Required visual reference package

Before production expansion, create and obtain approval for a creature turnaround, three compatible creature variants, a habitat composition, a material sphere sheet, a scale lineup, a lighting test, and one in-engine benchmark frame. Include neutral clay renders alongside beauty renders so surface quality is distinguishable from lighting.

Compare at the actual gameplay camera, resolution, and target hardware. Score silhouette, proportions, expressiveness, material cohesion, environment density, readability, and technical performance. Passing requires owner approval of the benchmark scene; numerical scores alone cannot authorize a new art style.

Blender is mandatory

Every production 3D mesh, creature part, prop, plant, terrain surface asset, structure, vehicle, rig, and authored animation must have an editable Blender source of record. Final geometry, UVs, normals, materials, rigging, and export preparation happen in Blender. Preserve linked dependencies or pack them reliably.

External reference imagery is not a production 3D asset. Any externally sourced or generated 3D input requires explicit approval and must still be authored and finalized in Blender. Merely opening an unchanged generated mesh does not satisfy authorship. Runtime terrain assembly, mesh deformation, and procedural placement require an approved boundary defining which Blender-authored sources they use. Engine-native collision shapes, navigation data, particles, and UI are functional systems; they do not excuse replacement of production graphical meshes with primitives.

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16 Asset production workflow

1. Write an asset brief with gameplay function, dimensions, camera distance, silhouette, rig family, material family, proposed budget, and acceptance criteria.
2. Build a Blender blockout and inspect it at gameplay scale. Confirm function before detail.
3. Author clean topology and UVs. Resolve shading, normals, transforms, origins, and attachment sockets.
4. Create materials and textures consistent with the approved benchmark. Record texture color spaces and packing.
5. Rig and animate where needed. Verify deformation at extreme supported proportions.
6. Create collision proxies and LODs when justified. Check that LOD changes preserve the silhouette.
7. Export to an approved interchange format, initially proposed as GLB. Check animation names, scale, materials, and sockets in Godot.
8. Produce a validation report and comparison captures in the benchmark scene. Admit only reviewed assets into production folders.

Start with one meter as one Godot unit and explicitly test Blender-to-engine orientation. Do not rely on remembered exporter conventions. Pin Blender, exporter settings, and Godot versions after a reproducible import test.

Initial profiling hypotheses: ordinary creatures 5,000–15,000 triangles; prominent creatures 15,000–30,000; small props 100–2,000; common textures 512–2,048 pixels. These are proposals, not guarantees or quality targets. Measure draw calls, skinning, overdraw, memory, and simultaneous actor counts before adopting budgets.

Each asset record includes stable ID, owner, source blend path, export path, dependencies, revision, dimensions, triangle counts, texture sizes, rig family, animation list, provenance, review state, and test results. Missing Blender source blocks production admission.

17 Godot architecture proposal

Pin a stable Godot 4 release only after checking the repository and available toolchain. Do not claim a specific minor version is installed without inspection. Prefer a desktop rendering configuration that meets the approved Windows performance target, with a fallback only after a deliberate review.

Separate domain rules from scene presentation. Genome validation, phenotype derivation, progression gates, resource transactions, and save migration should be testable without loading the whole world. Scenes compose views, input, movement, and interaction adapters.

System	Owns	Must not own
Genome service	Validation and derived phenotype	UI layout or save file access
Interaction service	Eligibility and action transactions	Camera behavior
Ecology service	Regional resources and populations	Per-frame animation
Era coordinator	Transition checks and state changes	Recreating lineage history
Persistence service	Versioned serialization and migration	Gameplay balancing
Presentation layer	Feedback, animation, camera, UI	Duplicate authoritative rules

Proposed directories: `game/`, `game/domain/`, `game/systems/`, `game/scenes/`, `game/ui/`, `game/data/`, `assets/source/blender/`, `assets/export/`, `tests/`, `tools/`, `docs/`, and `builds/`. Keep imported caches and generated build products out of source control. Use large-file storage for substantial binary sources where supported.

Use stable IDs and versioned data schemas. Avoid string-based scene-tree lookups for critical contracts. Keep global services few and explicit. A new abstraction must solve a current problem; later eras do not justify speculative frameworks before the first playable loop.

18 Persistence and world continuity

A save contains schema version, build identifier, world seed, lineage ID, genome, generation, current era, discoveries, relationships, settlement records, world changes, player location, and active transactions or safe equivalents. Never serialize arbitrary executable code.

Write to a temporary file, validate, then replace the active save using the platform's supported safe mechanism. Retain a last-known-good backup. Protect saves from interrupted writes and failed migrations. Show a useful error and preserve originals when loading fails.

Test every approved schema migration with fixtures. Stable IDs survive renamed files. Separate account settings from run state. Era changes are transactions with pre-transition checkpoints and explicit migration tests. Loading a save must never silently remove body parts or erase lineage history.

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19 Controls and accessibility

Support remappable keyboard and mouse controls first, with controller support planned explicitly. Display inputs from bindings rather than hardcoded labels. Provide adjustable camera sensitivity, invert options, readable text scaling, subtitles, reduced camera motion, and settings for flashing effects.

Use shape, animation, labels, or audio alongside color. Critical information must not rely on hearing, color discrimination, or precise repeated button presses alone. Tooltips explain anatomy tradeoffs with plain language and comparable values.

Separate exploration, editor, settlement, and strategic input contexts. Opening a menu must not leave a dangerous gameplay action held. Tutorial prompts appear when relevant and can be dismissed or revisited. Retain a concise objective and discovery journal without prescribing one optimal evolutionary path.

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20 Audio and presentation

Create original creature vocal families linked to size, mood, and communication intent. Use environmental sound to establish habitats and communicate nearby activity. Provide visual equivalents for critical cues. Music expands with the player's scale while retaining recognizable motifs.

Animation states communicate attention, fear, effort, satisfaction, and social response. Blend locomotion and actions without allowing visual transitions to alter authoritative costs or damage. Audio and effects follow confirmed events rather than predicting unvalidated outcomes.

21 Agent operating procedure

Every agent begins by reading the current Bible, applicable repository instructions, approved task, relevant decision records, and affected tests. State the task boundary and identify which requirements it implements. Never assume a prior chat grants access to a machine, repository, API, or asset library.

Task brief template: task ID; requirement IDs; approved scope; allowed files; prohibited changes; dependencies; expected behavior; verification steps; required evidence; open questions; rollback method.

Execution workflow: inspect → identify the smallest coherent change → implement within scope → run relevant checks → exercise the player flow → capture evidence → report limitations → request review. Keep changes reviewable and preserve unrelated user work.

Agents must not alter era count, art pipeline, project separation, core continuity, public schemas, or agreed milestone scope without approval. Do not add paid services, runtime AI, network dependencies, or imported art packs as routine implementation choices. Credentials never enter source control or logs.

When blocked by an actual design decision, document options and recommend one. Continue independent authorized work. Do not invent approval, claim unseen tests passed, or describe a build as playable solely because the project parses.

Handoff template: behavior delivered; files changed; requirement coverage; checks actually run; build identifier and location; visual evidence; known issues; decisions needed; next bounded task. Label work as proposed, implemented, or verified accurately.

Parallel agents, when explicitly authorized, receive non-overlapping responsibilities and an integration owner. Shared schema changes require coordination. This document does not itself authorize spawning agents or consuming external API budgets.

22 Verification and quality gates

Meaningful automated coverage

- Genome validation rejects unsupported attachment graphs without spending resources.
- Phenotype derivation reflects approved benefits, costs, caps, and diminishing returns.
- Feeding and construction transactions cannot duplicate resources or charge failed actions.
- Era transitions retain lineage, anatomy, relationships, and discoveries.

- Save and load round-trip representative state; migration and corrupt-file recovery preserve originals.
- AI exits unreachable tasks and pursuit states; simulation fidelity changes do not duplicate entities.
- Asset validation catches absent Blender sources, missing dependencies, invalid scales, and broken exports.

Use unit tests for rules, integration tests for service contracts, and bounded scenario tests for actual play. Avoid tests that merely repeat constants or mirror implementation. Visual quality requires rendered review, not only assertions.

Playable Windows gate

Export a Windows build using the pinned export templates. Launch it on Windows or an explicitly documented compatible test environment; packaging alone is insufficient. Test fresh-run startup, movement, interaction, editor commit, save, exit, reload, death recovery, and one complete milestone loop. Check missing assets and runtime errors.

Record commit, engine version, export configuration, hardware, resolution, scenario, and observed failures. Test outside the editor. If Windows execution is unavailable, report **exported but not Windows-verified** and leave that gate open.

Visual gate

Review actual gameplay captures alongside the approved benchmark at equal camera scale and comparable lighting. Inspect creature identity, contact with terrain, animation, material consistency, target readability, and UI. Do not substitute generated beauty imagery for in-engine evidence.

Performance gate

Proposed target is 60 frames per second at 1920×1080 on a named baseline PC. Hardware and encounter density remain unapproved. Set these before claiming compliance. Measure CPU and GPU frame times, frame-time spikes, memory, loading, and representative worst-case scenes. Budgets must include UI, simulation, rendering, and asset streaming together.

23 Incremental production roadmap

Milestone	Deliverable	Exit evidence
M0 Foundations	Separate project, pinned toolchain, source rules, save schema	Reproducible launch and validated sample Blender export
M1 First life	Microscopic feeding, threats, reproduction, bounded editor	Two-generation playable Windows loop and rule tests
M2 Visual benchmark	Cohesive creature and habitat set	Owner-approved in-engine comparison frames
M3 Anatomy depth	Multiple viable body variants and reliable locomotion	Same-scenario comparisons with meaningful tradeoffs
M4 First continuity	Approved aquatic or land transition slice	Before-and-after save and gameplay continuity evidence
M5 Social foundation	Companion identity, communication, group task	Persistent relationship scenario and recovery checks
M6 Settlement slice	One functioning settlement and neighbor	Resource, construction, diplomacy, and embodied play loop
M7 Regional slice	Connected settlements and routes	Regional consequence and local interaction scenario
M8 Space slice	Second world, outpost, return home	Cross-world persistence and homeworld relevance
M9 Production readiness	Content completion and quality closure	Agreed test matrix, Windows build, asset provenance audit

This is dependency order, not a schedule or claim of completed work. Avoid mass asset generation before the benchmark passes. Expand one validated loop at a time. Later milestones may refine estimates after earlier risks are measured.

24 First playable production brief

Proposed scope: one compact microscopic habitat, one player rig family, three body configurations, two food types, two threat behaviors, one reproduction interaction, a bounded anatomy editor, and a save slot with backup recovery. Counts are scope caps pending approval.

The player can start, feed, encounter danger, reproduce, change one functional body trait, observe its effect, save, reload, and repeat. The habitat contains enough regeneration to recover from mistakes. A debug panel exposes current genome ID, energy, derived movement, AI state, and seed in development builds only.

Deliver Blender sources and exports for every production graphical asset. Temporary placeholders must be visibly marked in the task inventory and removed before visual acceptance. Provide one Windows build, a concise player walkthrough, automated test results, and real gameplay captures. Do not include later-era mock buttons that imply implemented functionality.

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25 Risk register

Risk	Early warning	Response
Arbitrary anatomy breaks animation	Foot sliding and invalid poses	Bound rig families and test extreme proportions early
Seven eras become separate games	Resets and unused inherited traits	Enforce continuity contracts at every transition
Art loses cohesion	Inconsistent materials and silhouettes	Approve benchmark before expanding asset volume
Simulation grows without limits	Frame spikes and population explosions	Fidelity tiers, caps, conservation checks, profiling
Agents drift from the vision	New dependencies and unrelated features	Scoped tasks, requirement mapping, review gates
Architecture grows faster than play	Many services without a complete loop	Prioritize first playable and current requirements
Builds work only in editor	Missing imports or export-only failures	Export and launch early on Windows
Earlier Bible conflicts emerge	Recovered requirements disagree	Compare explicitly and obtain owner resolution

26 Decisions requiring owner review

The confirmed requirements remain binding regardless of these open items. Review the seven proposed era names and boundaries; approve the first playable scope; choose supported starting body plans; approve camera and movement constraints; establish the visual reference package; name baseline Windows hardware; confirm difficulty and death recovery; and choose repository destination and toolchain versions after inspection.

Additional design decisions include how long players may remain in an era, viable non-terrestrial routes, the scale of strategic abstraction, conflict intensity, spacecraft customization, and the balance of embodied and management play. These should be resolved before their dependent milestone, not invented by an agent during implementation.

27 Definition of done

A feature is complete only when it satisfies its approved behavior, integrates with the actual player loop, preserves relevant prior systems, includes Blender sources for graphical assets, passes appropriate checks, has honest execution evidence, and is recorded in a reviewable build. Known limitations must be visible in the handoff.

An era is complete only when its activities are enjoyable and understandable, more than one approved evolutionary approach is viable, its transition contract passes, saves remain intact, visual quality meets the benchmark, and the Windows acceptance scenario works outside the editor. No agent can declare the entire game finished from a checklist of unconnected demos.

28 Change control

Each approved revision records date, decision, reason, impacted requirements, migration implications, and owner approval. Retain superseded specifications for traceability but clearly identify the current version. Proposal approval should be recorded explicitly before promoting it into binding production requirements.

Before beginning work, an agent should be able to answer: What player behavior am I delivering? Which approved requirement authorizes it? Which inherited system must remain intact? Which source assets are required? What evidence will prove it works? If these answers are missing, resolve the relevant task brief rather than improvising the vision.